

Operation manual for RHEED monitoring Software for RHEED image processing



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For PCI-1410, without Scanner

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All product names, such as *Windows*, *MS-Excel*, *Pentium*, *HASP* mentioned in this manual are trademarks or registered trademarks of the respective companies.

This manual assumes the users know well about the operation of *Windows* system.

1. Outline

The purposes of this program are acquiring the RHEED patterns on the fluorescence screen through the CCD camera, and monitoring the vibration of the RHEED intensity. Also the following functions are available;

Display, FFT analysis, Data storing of the RHEED intensity vibrations.

Display of RHEED pattern as the video mode and their storing.

Readout of the stored data, modifications and storing as the text mode.

(for analysis with some Spread Sheet software like *MS-Excel*)

Measuring of the lattice constant from the RHEED pattern.

Profile extraction by the Gaussian Fitting.

Suppressing the influence of the stray light, *etc.*

You can perform the deposition process utilizing the RHEED system much preciously and efficiently with this program.

2. Working Environment

This program requires the following hardware and software environments;

An IBM PC compatible computer

CPU: Intel Pentium-4 processor, with the CPU clock higher than 2 GHz

One PCI bus slot (Full height)

OS: Windows2000 SP4, or XP

more than 512 MB Memories (not certified below that, recommended more then 1 GB)

HDD rooms more than 2 GB (10 GB is recommended)

Graphic Card with 128 MB memories is preferable (not required)

Picture Processing Card: any of bellows from National Instruments Inc.

PCI-1410, PCI-1407(discontinued), PCI-1408(discontinued),

PCI-1409(discontinued)

Control software

RHHED monitoring software

IMAQ Runtime License from National Instruments *Inc.*

Other software drivers

Hardware Protect Key device (HASP: shipped with the system)

Video camera set (monochrome) EIA, NTSC, CCIR, PAL, RS-170 are available

SONY XC-75(CCD camera), Power Supply, Zooming lens and Cables are

Included in the standard set.

3. Operations

3-1 Preparation

Connect the CCD camera to the PC and turn ON the power supply unit of the camera.

3-2 Start the RHEED monitor.

Double-click the "RHEED monitor" icon  on the desktop.

Following three windows will appear (Fig.1 (b) is just an example, an image from CCD camera will be displayed.)

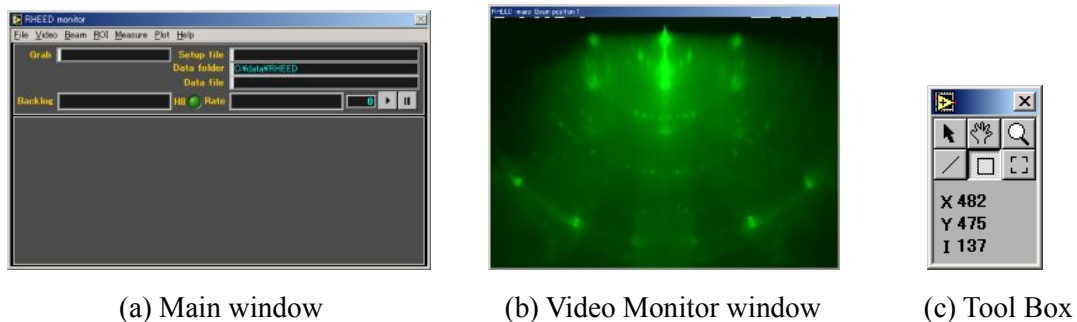


Figure 1 Three windows at start.

The Window size of the Video Monitor is 640x480 pixels.

Figure 2 shows the Menu-Bar on the Main window.



Fig.2 Menu bar of the main window

Followings are the functions that can be selected from the Main window.

[File] :

[New Sample] Specifies the folders for data saving.

The default is "C:\¥data".

Please find details in the Section 3-7 "To specify the data folder".

[Start recording] Starts the data acquisition.

The same function as clicking the  button on the Main window.

[Stop recording] Stops the data acquisition.

The same function as clicking the  button on the Main window.

[Pause recording] Momentary stops the data acquisition.

The same function as clicking the  button on the Main window.

[Parameter list] Selects the data to save.

Please find details in the Section 3-8 "To acquire and save the data".

[Extract data] Extracts the saved data.

Please find details in the Section 3-10 "To display and save data".

[Load Setup] Loads the setup file for Scanning functions.

Not used without the beam scanner.

[Save Setup As] Saves the setup file for scanning functions.

Not used without the beam scanner.

[Edit] Momentary stops this program.

Click the  button to restart the program.

[Exit] Quits this program.

[Video] :

[Show image] Activates the Video Monitor window.

The hidden Video Monitor window will be popped up.

[Show overlay] Selects whether or not to show the ROI (Region Of Interest) selected area for intensity measurement) in the Video Monitor.

ROI will be displayed in case this command is checked.

[Edit color map] Sets the color map of the Video Monitor.

Please find details in the Section 3-11 "To edit the color map".

[Display settings] Setups the Video Monitor settings.

Please find details in the Section 3-12 "To setup the Video Monitor".

[Video settings] Tunes the Black level (brightness) and White level (Contrast) of the video signal.

Please find details in the Section 3-13 "To tune the video input signal".

[Grab image] Saves the image on the Video Monitor.

[Grab setup] command should be used beforehand to setup the file format.

Please find details in the Section 3-8 "To acquire and save the data(2); to acquire the data".

[Grab setup] Setups the image file format used in the [Grab image] command.

Please find details in the Section 3-8 "To acquire and save the data(1); to save the data".

[Video source] Selects the video input. Set "A free-running camera" mode without the scanning function.

Please find details in the Section 3-13 "To select the video input".

[Beam] : This menu lists commands concerning to the scanning functions.

Not used in case without scanning functions.

[ROI] : (Commands for setting up the monitoring region. ROI : Region Of Interest)

[Cut all] Deletes all the ROI.

[Cut] Deletes the activated ROI. The activated ROI is shown at the ROI button  in the Main window, and also indicated with the red frame on the Video Monitor (light blue frames for other ROIs).

[Copy] Copies the activated ROI (like the copy command on Windows).

[Copy all] copies all ROIs.

[Paste] Pastes the copied ROI (like the past command on Windows)

[Shift] Opens “Move ROIs” windows, and Moves ROIs all at once.

Moves 1 pixel by the one-arrow button toward the arrow, 10 pixels by two- arrows button.

!Notes : in case no ROI is activated, these [ROI] menus are unavailable.

[Measure] : (Setting up the measuring parameters for activated ROI. Details are listed in the section 3-5 "Detailed settings for monitoring region".)

[Remove] Removes currently activated parameters.

[Min and Max] Acquires the Max/Min intensities and the position data in the ROI.

[Average] Gets the average intensity in the ROI.

[Centereid] Gets position of the center of the ROI.

[Fit Gaussian] Fits the intensity distribution in the ROI by the Gaussian distribution (the normal distribution) patterns, and gets the relational parameters.

[Distance] Gets distance of the two ROIs.

[Plot] : (Opens the Real Time Monitors window. Details are described in the section 3-6 "Real Time Monitor").

[Parameters] Opens the Real Time Monitor window for the intensity chart.

[Line sections] Opens the Real Time Monitor window for the intensity pattern on an established line.

[Data Viewer] Displays the saved data, or opens window for FFT analysis.

This document follows the measuring procedures hereafter;

3-3 Adjusting the Video Input (Video Settings)

The first thing to be done is adjustments of the video input.

This procedure is (1) the initial setting of the Video Card, (2) adjustments of the iris and the focus of the CCD camera (lens), followed with the fine adjustments.

(1) Rough adjustment of the brightness and contrast of the Video card

Click from the Menu bar of the Main windows into [Video→Video settings] and get the followings
Fig. 3 Video Setting window;

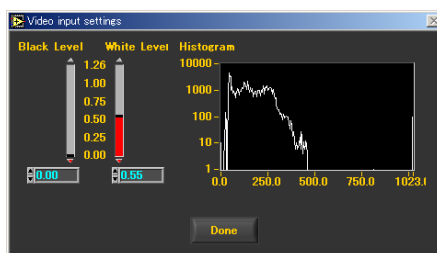


Fig.3 Video Setting window

Put the values, [0.00], [0.55] into [Black level], [White level], respectively

(Black level: Brightness, White level: Contrast)

Click  button to close the window after entering the values.

(2) Rough adjustments of the contrast and focus of the video camera

Turn the ring on the camera lens to adjust the contrast and the focus.

Functions of the rings are as shown in the next Fig. 4.

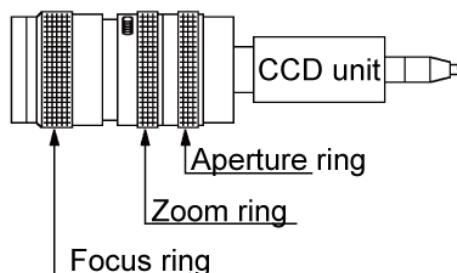


Fig. 4 Rings on the lens of the video camera

Adjustments of the camera lens are as follows;

1. Get the electron beam spot on the fluorescence screen using the RHEED controller.
2. Turn the zooming ring to get the fluorescence screen on the whole window.
3. Turn the focus ring to make the beam spot clear.
4. Turn the iris ring to adjust the contrast.

Then decrease the filament current to reduce the spot brightness barely seen by naked eyes on the screen, and adjust the iris ring to get the beam spot on the window.

Increase the filament current to make the spot brighter and confirm if the beam spot on the window is not saturated. Stop the diaphragm in case the beam spot on the window is saturated, or open the diaphragm if the spot is gloomy. Repeat these to get the optimum condition (dynamic range adjusting).

(3) Fine adjustment of the Video Car

Adjustments of the Video Card must be performed with the Video Setting window

shown in Fig.3, again.

Adjust finely the values of [Black level], [White level] on the window to get the maximized dynamic range.

If you already have the enough dynamic range just after the camera lens adjusting, there is no need to fine adjustment of the Video Card.

It would be difficult to get the optimum dynamic range only by the adjustment of the Video Card or Camera Lens, and then there would be necessary to repeat the fine adjustment of them.

Since the brightness of the beam spot depends on the material of the substrate of the target, you have to adjust the brightness before every deposition.

3-4 Setting of the Video Monitoring Region

Select a tool for setting the monitoring region from the Tool Box window shown in Fig. 5.

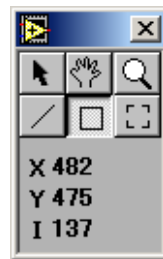


Fig. 5 Tool Box window

The tool to set the rectangular area (box) is selected as the default.

Click  if you want to make a line.

Meanings of the tools in the Tool Box are as follows;



: Changes the size of the ROI. Click and activate the ROI you want to change the size, then drag it with the mouse to change the size easily.



: Enhance or shrink the window size. Click to enhance the window and click with pressing the Shift key to shrink.



: Moves the enhanced window. The window moves with the mouse dragging.



: Currently unavailable.

Current position and the intensity at the cursor are displayed beneath the tools.

Set the monitoring areas in the Video Monitoring window by the mouse dragging with the selected tool.

Figure 6 shows an example that 6 boxes are set in the monitoring window.



Fig. 6 Example of ROI setting (6 boxes)

Once a box or a line is settled, the main window shows the monitoring information for the ROI as shown in Fig. 7.

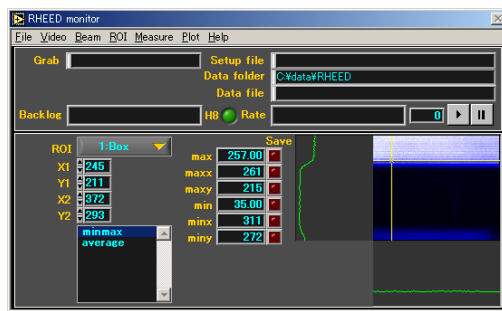


Fig. 7 the Main window showing the ROI

3-5 Detailed setting of Monitoring Region

The following parameters can be set for the settled ROI. These parameters can be displayed in the Real Time Monitor or saved into the log file.

The following operations are available for the current activated ROI.

You can confirm the activated ROI with the ROI button **ROI** **1-Box** in the Main window, and also the ROI is indicated with the red frame in the Video Monitoring window (the other ROIs are shown by the light blue frames).

You can select parameter items in the pull down menu appeared when the [Measure] tab in the menu bar of the Main window.

Details for menu items and for parameters as follows;

[Remove] Removes parameter currently activated.

Activated parameters are shown in light blue in the Parameter List box in the Main window.



In this case, [Min and Max] are activated.

[Min and Max] Adds the following parameters in the intensity data for the activated ROI.

- Maximum intensity (Max)
- X coordinate of the maximum intensity (maxx)
- Y coordinate of the maximum intensity (maxy)
- Minimum intensity (Min)
- X coordinate of the minimum intensity (minx)
- Y coordinate of the minimum intensity (miny)

There are default values for these items ([pixel] for the coordinates)

[Average] Adds the average intensity (ave) in the activated ROI as the parameter.

This item is set as the default.

[Centroid] Adds the following parameters of the center position for the activated ROI.

- X coordinate (centx)
- Y coordinate (centy) [pixel] for both

[Fit Gaussian] Calculates the Gaussian (normal) distribution fitting for the intensity distribution in the activated ROI.

By the fitting calculations, you can have the intensity distribution profile as the Gaussian parameters, or can eliminate the background of the stray light.

When the intensity distribution might not of a clear spot, the fitting calculation might fail to express the intensity as the Gaussian distribution, and result in the meaningless parameters. In such case, ignore (or delete) these parameters.

Following parameters are added by the fitting calculation.

(for Box ROI)

- Peak values after the fitting (X-axis : amp, Y-axis : Yamp)
- Position where the peak value is obtained (X-axis : Xpos, Y-axis : Ypos)
- FWHM of the intensity peak(X-axis : Xwid, Y-axis : Ywid)
- Background intensity (X-axis : Xbck, Y-axis : Ybck)
(Background : the offset value of the intensity)

(for liner ROI)

- Peak value after the fitting (Amp)
- Position from the end of line (X1,Y1) where the peak is obtained (Pos)
- HWHM of the intensity peak (Wid)
- Background intensity(Bck)
- Absolute position of the peak (X-axis : Posx, Y-axis : Posy)

Followings are required parameters for fitting calculation.

- Iteration loop count (Loop) default : 10
- Start point of the line (no need for Box ROI) (From)
- End point of the line (no need for Box ROI) (To)

“From”, “To” can be mouse-dragged.

For the Box ROI, Parameter List Box shows "2dfit" and "fitgaus" for Liner ROI.

Multiple fitting zones can be set for the Liner ROI, and Parameter List Box shows "fitgaus_1", "fitgaus_2" and so on.

When multiple zones are set on the same line, change of the lattice constant can be measured from distance variation of the streak patterns using [Distance] parameter.

[Distance] Adds the parameters for measuring the distance between the activated ROI and another ROI selected by the user.

- X-axis distance (Dx)
- Y-axis distance (Dy)
- Distance in a straight line (Dxy) (all in [pixel])

Targets for distance calculation are one of the followings;

- Max (the brightest position in the ROI)
- Min (the most dim position in the ROI)
- Cent (the center position of the ROI, in case Centroid parameter is added)
- Fit (the peak position of the Gaussian distribution after fitting calculation)

3-6 Real Time Monitoring

!!Attention!! Data cannot be saved in the Real Time Monitoring. Follow the procedures described in section 3-7 "Set of the data folder" and 3-8 "Acquiring and saving data", if you want to save the data.

For the Real Time Monitoring of the RHEED oscillation, following three windows are prepared.

(1) Parameter window

Cricking [Plot→Parameters] on the Menu bar in the Main window shows the Parameter window (Fig. 8); This window shows the parameters for each ROI.

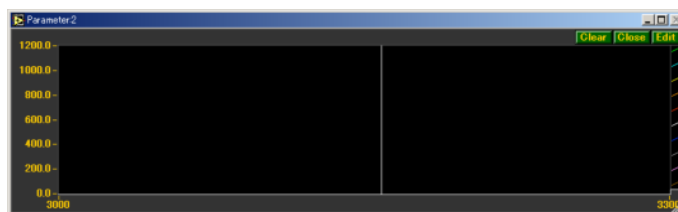


Fig. 8 parameter window

Cricking **Edit** button on the upper-right shows the entering area  for the number of the ROI. Crick  (Left: ROI, Right: parameter) button to select the monitoring parameter. The horizontal and vertical axis are time and intensity, respectively.

The default value of the redrawing time (X axis) is 300 (sec) and can be increased to 500 (sec). To change the time scale, set the "PlotHistory" value in the "RHEED.Ini" file in the value range between 1 and 500. (This change will be effective on the next execution of the program)

The vertical axis of the intensity is 0 to 1200, as the default, and can be changed any time by selecting the maximum or minimum with mouse-cricking. The maximum value of the intensity is 1024 (corresponding to 10 bits).

Clear button resets the previous data and starts to new plotting.

Close button closes the window.

Figure 9 shows an example of the intensity monitoring.

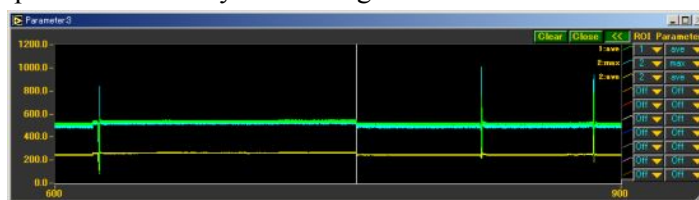


Fig. 9 an example of the intensity monitoring.

There is no limitation excepting the CPU memories to open a new parameter window. The window size can be changed freely by mouse-dragging at the lower-right corner of the window.

(2) Line section window

Cricking [Plot→Line section] on the Menu bar in the Main window shows the Line section windows shown in Fig. 10.

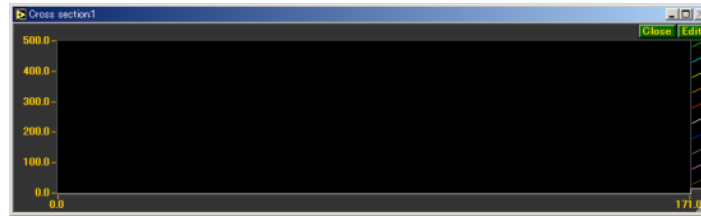


Fig. 10 Line section window

The Line section window displays the intensity pattern graph on the line which defined by the  tool in the Tool box.

The horizontal axis of the Line section window is the coordinate axis (liner distance), while the time in the Parameter window.

The **Edit** button can select the line or parameters as in the Parameter window.

Selectable parameters are the "profile" (raw data) and "fit" (fitted data by the Gaussian distribution).

Figure 11 shows the example for selecting line and the parameter in the Line section window.

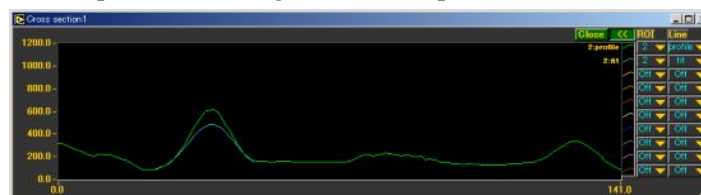


Fig. 11 Example of monitoring in the Line section window

There is no limitation excepting the CPU memories to open a new Line section window. The window size can be changed freely by mouse-dragging at the lower-right corner of the window.

(3) Data Viewer window

Cricking [Plot→Data Viewer] on the Menu bar in the Main window shows the Data Viewer window shown in Fig. 12.



Fig. 12 Data Viewer window

The Data Viewer window is a tool to display or to analyze the logged data saved in the file. This tool requires the procedure described in the section 3-7 "Data acquisition and saving", beforehand, to display or to analyze the data in the course of saving (or reading) the data.

During the data acquisition, the data file name, including the folder name, will be displayed in the "Data File" section  `D:\data\RHED\2008_07_17\test_080717_1.dat`.

Parameters displayed in above the Plot window will be shown in the area  `1:1:Box:average:ave`. The parameters can be selected in the pull-down menu shown after cricking. Parameters can be also changed by cricking the arrow   button.

When the "Hold" **Hold** button is clearly seen (non-hold state), the data graph shows from the start of the data acquisition and to the present time, redrawing the data repeatedly. While the "Hold" **Hold** button is dimmed (Hold state), graph refreshing is stopped. In the "Hold" mode, after selecting the region with the "Start" cursor and "End" cursor, by cricking the "FFT" **FFT** button, RHEED oscillation frequency can be obtained.

Detailed explanation can be found in the section 3-10 "Display and Saving the logged data".

The **Done** button closes the window.

Figure 13 shows an example showing the data display and analysis.

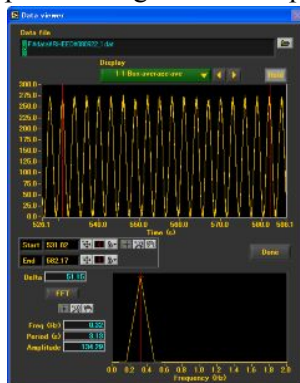


Fig. 13 Example of data display and FFT analysis in the Data View window

3-7 Data Folders

Various RHEED parameters and data are saved in the folder defined in this section.

Cricking [File→New Sample] on the Menu bar in the main window shows the Folder Setting window as shown in Fig. 14.



Fig. 14 Folder Setting window

The section [Path] shows the current folder name.

Double cricking the [<Parent folder>] shows the upper parent folder name.

The "New" **New** button creates a new folder.

The section **Data folder** **C:\data** at the upper-right corner in the Main window shows the current data folder name.

3-8 Data Acquisition and Saving

The intensity data acquisition and saving performed with following procedures;

(1) Selection of the data field, (2) Data acquisition.

(1) Selection of the data field

There are following two ways to select the data field.

[I]. To select data at once from the Parameter List window.

Cricking [File→Parameter list] on the Menu bar in the Main windows shows the Parameter List window shown in Fig. 15.

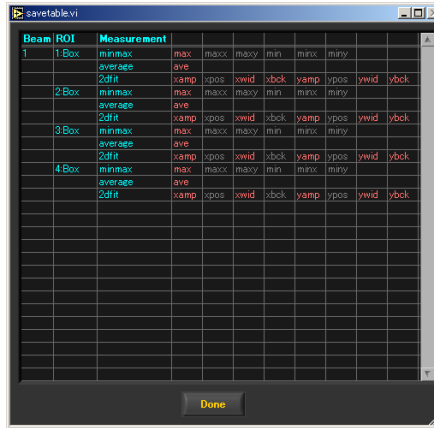


Fig. 15 Parameter List window

This window summarizes the all selectable parameters classified with the Box and the Line. Move the cursor onto the parameter you want to save as a logged data, and crick to select it. The parameters indicated with red color will be saved, and ones with gray color will not. The "Done"  button closes the windows.

[II] To select individually from the ROI information.

As shown in Fig. 16, there are lamp switches on the right of the parameter values for each ROI in the main window ("Save" is seen on the top). Cricking the lamp switch selects the parameter to be saved (In Fig. 16, Xamp and Yamp are turned on and selected).



Fig. 16 Lamp Switches to select saved parameters

These lamp switches in the ROI and the list in the Parameter List windows correspond with each other, the selected one by both methods reflects on each other.

(2) Data acquisition

Cricking the  button in the main window, or cricking [File->Start Recording]on the Menu bar in the Main window show the File Name window shown in Fig. 17.

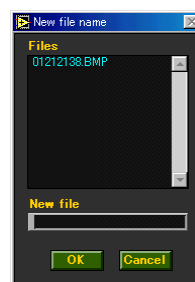


Fig. 17 File Name window

Enter the file name in to the area "New file" , then crick  button to start the data acquisition.

The button  stops acquisition and the button  pauses it.

The default folder to save the data is [C:\¥data].

To change the data folder, please refer the section 3-7 "Setting of the data folder".

3-9 Saving the Graphic Images

Graphical image displayed in the Video Monitor window can be saved as the still picture. Procedures are setting the file format, the image resolution followed with the image acquisition according with the setting.

There is no need to setup the format or the resolution every time, if you have already set these up.

(1) Setting up the saving file

Cricking [Video→Grab setup] on the Menu bar in the Main window shows the Set Up windows as shown in Fig. 18.

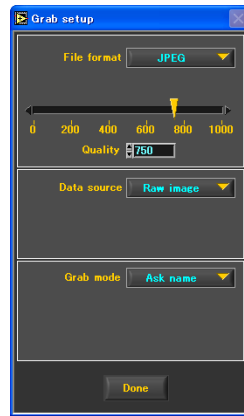


Fig. 18 Set Up window for image storing

Set up items are as follows;

[I] File format

Cricking "File format"  selector makes you to choose the image file format in the followings. The default is JPEG.

BMP, PNG, TIFF, AIPD, JPEG, Text, Binary

[II] Compression/Resolution

When BMP or TIFF is selected as the file format, you can chose whether or not to compress the data and compression format.

For PNG and PEG, the image resolution can be selected. You can choose the resolution by sliding the "Quality"  bar, or by entering the resolution value directly. (unit : [dpi])

[III] Selecting the image specification

By cricking "Data source"  selector, the image specification can be selected in one of the bellows.

Raw image, Average, Display gray

Specifications for the above are as follows;

Raw image: Image data as the input into the Image Processing Board

Resolution depends on the board.

(ex : PCI-1407=8bit, PCI-1410=10bit)

Average : After averaging processed.

Display gray: Compressed into 8 bits.

In some cases, the error message as shown in Fig. 19 may appears showing the selected data defined in the "Data source" menu does not meet the file format.

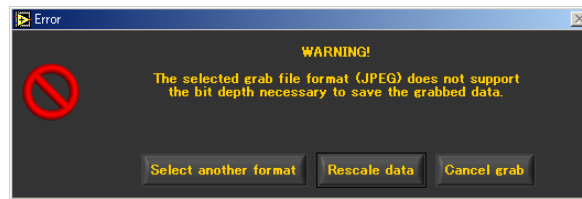


Fig. 19 Error messages in the saving image data

[IV] Selection of displaying ROI

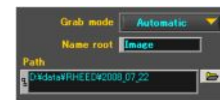
When "Display gray" are selected in the selection of the saving image, the "ROI overlay"   switch appears. You can choose whether or not to display the information on the Box and Line set in the Video Monitoring window onto the Image saving data.

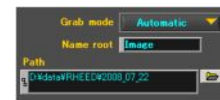
Selectable items are one of "Discard": Not displayed, or "Save": Display, the default is "Discard".

[V] Selection of asking the file name

By cricking "Grab mode"  , you can chose whether or not to be asked the file name at saving the image data.

"Ask name: Asked, "Automatic": Save immediately (no asking)



When "Automatic is selected, the window changes to , you can define the data folder name and the basic file name. Every time you crick the "Grab image", the still image data will be saved into the basic filename followed with a series number such as "0001", 0002. It would be convenient in saving the image transitions.

[VI] The  button closes the Set Up window.

(2) Storing the image

Cricking [Video→Grab image] on the Menu bar in the Main window, the current displayed image is stored as the still graphic data. If the "Ask name" was selected in the sub-section [V] of the above (1), a window asking the file name will be popped up, or for the "Automatic mode, the image will be immediately.

The stored graphic data can be displayed or printed by any graphic editing software. This program has no function of treating the graphical data.

3-10 Display and Store of the Logged Data

The data stored as described in the section 3-8 can be displayed, modified or stored in the text mode.

Cricking [File→Extract data] on the Menu bar in the main window shows the Data Read Out window as shown in Fig. 20.



Fig. 20 Data Read Out window

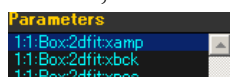
The file name of the data whose graphics is displayed in the window is indicated in the section "Data file". During the data saving, the current saving data will be opened automatically.

To open a saved file, click  button and select file name.

The parameter data displayed in the graphic window is a selected one by the "Display"  button.

To change the displayed parameter, click to select on  selector, or  button.

The whole displayed data, or any data region can be stored in the text mode. The parameters to be saved are selected in the section "Parameters", and the selected ones are indicated with blue characters as the next figure.



In this case the upper one is selected.

Multiple parameters can be selected by clicking with the Shift key pressed.

"Select all" button  can select all parameters at once indicated in the "Parameters section.

The following procedure describes how to define a required region (time axis) and to save into the file. The region is defined with the cursors "Start" and "End".



The cursor operation uses tools in the  section.

Numerical values can be entered directly into the "Start" and "End" sections of the  section. (Column name of "Start" and "End" can be changeable.)

The icon  indicates which is selected in "Start" and "End" with a tiny green lamp. In case the two lines lies on each other, clicking this icon to define one of the two, to move the selected line.

Clicking  icon show the following pull-down menu.

- Color: Changes the color of cursor.
- Cursor Style: Selects a style of the cursor.
- Point Style: Selects a figure of the point on the cursor.
(The point might be located at the outside of displayable area.)
- Line Style: Selects the cursor line style.
- Line Width: Selects the thickness of the cursor line.
- Show Name: Displays names of lines in case this box is checked.
Defaults are "Start" and "End", and they are changeable.
- Bring to Center: Moves the cursor and the point at the center of the graph
- Go to Cursor: Re-draws the graph as the current cursor at the center of the graph.

The icon , the Lock Menu icon show the following pull down menu.

- Free: Makes the cursor free on the graph.
- Snap to point: Moves cursor at the most closest data point on the graph, therefore the cursor moves on some of the data lines.
- Lock to plot: same as the "Snap to point", but for one data line.

The icon  moves cursor, and is effective when the color is thick.

Set the mouse pointer on the cursor and drag it.

The icon  is a tool to enhance and shrink the graph, and effective for a thicker color.

- Zooming of the defined graph area
- Enhancing the defined graph area to X-axis direction
- Enhancing the defined graph area to Y-axis direction
- Re-draw whole area
- Enhancing around the mouse cursor
- Shrinking around the mouse cursor

The icon  is a tool to move whole graph area freely, and effective for a thicker color.

To change the display size of the graph area, crick the Max/Min figures of the X and Y axes, and enter the numerical values.

The "Save"  button saves data into the specifies file.

The "Done"  button closes this menu.

3-11 Editing the Color Maps

This program displays pseudocolored image on the Video Monitoring window from the input of the monochrome CCD camera signal. The displayed color can be matched with the fluorescence color on the fluorescent screen. The default is green as the ordinary fluorescence color.

This color can be changed freely as follows;

Cricking [Video→Edit colormap] on the Menu bar on the Main window shows the Color map (Fig. 21).

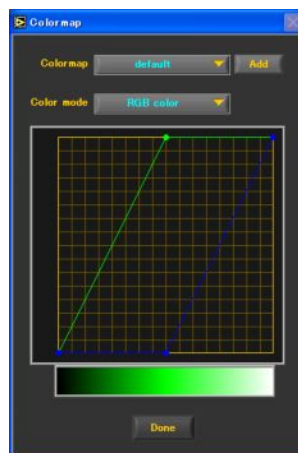


Fig. 21 Color Map window

Cricking the "Add"  button on the upper-right of the window asks you about the new color map name. Enter the name, then the window changes as bellow (Fig.22).

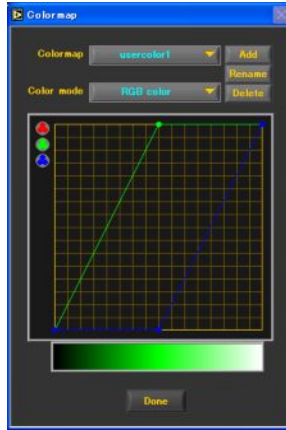


Fig. 22 User defined Color map window

By clicking "Color mode" **Color mode** **RGB color** button, you can select the display mode.

"RGB color": Changes the ratio of Red, Green and Blue.

"Grayscale": Changes the ratio of Black and White.

To modify the color map, drag the circular dot in the chart, the color change reflects on the Video Monitoring window. There are three points as defaults, clicking the "Point Add"  button at the upper-left of the chart adds the color of the clicked point, and enables the fine color adjusting. (The color of the "add point" button corresponds with each color of RGB.)

Clicking "Add" button again makes another color map.

If you want to change the name of color map, click "Rename" **Rename** button, and to delete the color map, use "Delete" **Delete** button.

This operation affects only the displaying color on the Video Monitoring window, not affect the video signal itself.

3-12 Setup of Video Monitoring window

The image mode of the Video Monitoring window can be changed.

Clicking [Video→Display settings] on the Menu bar in the main window shows the Display setting windows as Fig. 23.

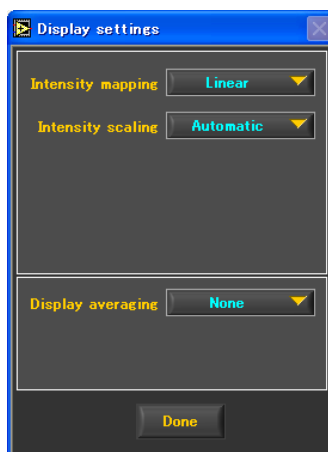


Fig. 23 Display Setting window

Selectable items are as follows;

[Intensity mapping] Sets the display mode.

[Linear] Linear mode

[logarithmic] Logarithmic mode. (effective in enhancing the large dynamic range, e.g. in observing a

- rather dim and small spot in the blight area.
- [Intensity scaling] Sets the brightness scaling
- [Automatic] Auto mode
- [Manual] Manual mode. Set brightness range manually.
Brightness range is defined by [Minimum] and [Maximum].
- [Display averaging] Sets the filter parameters for noise reduction.
Setting of this menu only affects the displaying window, not the intensity data.
- [None] Does nothing, displays the original image.
- [Add frames] Displays after add-averaging calculation for number of times set in the [Frames] menu.
"Multibeam scaling" should be used "Absolute" for the Non-Scanning mode.
- [IIR filter] Displays images after filter calculation of the IIR(Infinite Impulse Response). Set the Feedback value in the "Feedback".
"Multibeam scaling should be "Absolute" as same as the above.
- [Moving average] Displays images after the simple moving average.
Set the number of times value into the "Frames".
"Multibeam scaling should be "Absolute" as same as the above.
- [Beam table] used in the Scanning mode, not available in the Non-Scanning mode.

3-13 Video Input selection

Cricking [Video→Video source] on the Menu bar in the Main window shows the Video Input Select window (Fig. 24). (ordinary, (A) Non-Scanning mode)

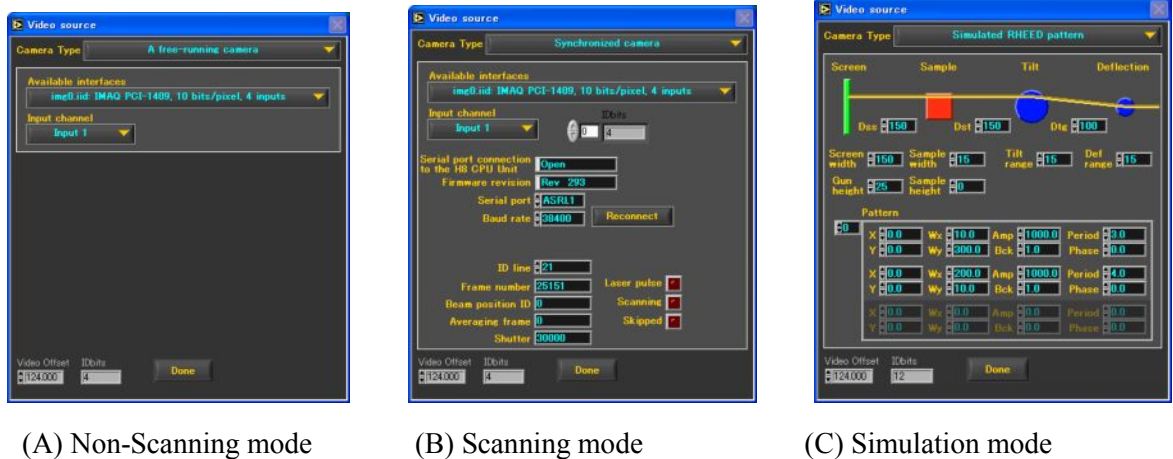


Fig. 24 Video Input Select window

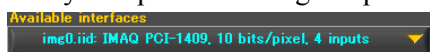
To change the mode, follow the pull down menu after cricking the "Camera Type" selector.



Followings are the correspondence between the mode and "Camera Type".

- (A) [A free-running camera]: Non-Scanning mode (used in this system)
- (B) [Synchronized camera] : Scanning mode
- (C) [Simulated RHEED pattern]: Simulation mode

If you replace the Image Capture board, crick the "Available Interfaces" button



, and select the available board listed in the menu. The following board can be used as of this manual's editing date.

PCI-1407 (Non-Scanning mode only), PCI-1408, PCI-1409, PCI-1410

For PCI-1408, -1409, -1410, set the input channel number by clicking the "Input channel" button. (the default is "Input 1")



This manual describes the operations for (A) [Non-Scanning mode], and the software was set up on this mode at the shipment. For any reason, if the mode was changed, follow the procedure of the above.

3-14 Program Termination

Click [File→Exit] on the Menu bar in the main window terminates this program. In case some windows remains, close them.

4. Troubleshooting

No	Trouble Matter	Causes and Solutions
1	“No Video Signal” is displayed on the Vide Monitoring window.	1. The video signal is not inputted. Check followings: [i] Are the Vide Camera and the Power Supply connected to PC correctly? [ii] Is the Camera power turned ON? 2. The camera input mode is invalid. Refer the section 3-13 “Selection of Video Input”, and check the “Camera Type”. Set the “Camera Type”to“A Free-running camera”
2	The program runs too slowly. The Image data acquisition also takes a long time.	Too many programs may run simultaneously. In case a red part in the [Back log] bar in the Main window flickering frequently, the load average of the CPU might be high. Close another window to reduce the CPU load.
3	The file size of the data file is 0 byte after recording the data.	1. Saving parameters are not defined. The parameters in the recording list must be defined before storing the data. If the parameters were not defined beforehand, nothing will be stored during the recording. Refer the section 3-8 “Data acquisition and Saving (1); selecting the storing parameters”, and check if the parameters are correctly set. 2. Data storage media There might be some issues in the storage media, such as HDD or another media if specified. Also please check if the enough room remains on the HDD or media to store the data.
4	The time axis (horizontal axis) of the Monitoring window cannot be changed.	While the maximum value of the intensity axis (the vertical axis) of the parameter window can be changed by mouse-clicking and setting the value, the time axis (the horizontal axis) are defined in the initial setting file. Details are described in the section 3-6 “Real Time Monitoring : (1) parameter window”.

No	Trouble Matter	Causes and Solutions
5	The parameter set in the Recording List disagrees with the recorded data.	If the ROI was changed after the Recording List was specified, data cannot be recorded properly. Do not change the ROI and parameters after setting the Recording List.
6	Another spreadsheet program cannot read the data file.	Since the data is stored in the binary mode, data format conversion to text file is necessary to use in the other software. Please refer the section 3-10 “Display and Save the logged data”, and read the logged data and save again in the text format.
7	This program cannot start with the error message “HASP not found”.	The Hardware Key Device [HASP] is inevitable to execute this program. In case this device is not connected on the PC, the error messages will displayed. When the message is displayed even the PC has the HASP, please contact the machine maker or the vendor.
8	The Video Monitoring window shows nothing and blanked.	1. The lens is all stopped down. 2. Contrast and Brightness are not properly set. Refer the section 3-3 “Tuning of the Video Input (Video Setting)”. 3. The lens cap remains on the lens. Another else there might be hardware issues, please contact the machine maker or the vendor.
9	The data monitoring is broken.	It might be affected by the Power Management of the Windows OS. In the property setting of the display in the Windows, set the “Power off” into OFF both for Monitor and HDD.

5. Inquiry

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